Table 1: Per-object contact fidelity and dynamic quality on CORE4D-G1 (both methods share the same 52 sequences / 11 objects). ↑: higher is better, ↓: lower is better. **Bold** = best. Ours = SPIDER-CEM. Contact metrics are penetration-aware (a loose "any-contact" count that rewards interpenetration is intentionally omitted). Dynamic-quality metrics for Ours and OmniRetarget are computed on a MuJoCo physics rollout (native for Ours; position-servo replay for the kinematic OmniRetarget, E197 protocol) via the same code, hence directly comparable.

		Phys. contact@3mm ↑		Phys. contact@10mm ↑		Phys. pen@3mm ↓		Phys. pen@10mm ↓		Max phys pen (mm) ↓		Geom. pen@2mm ↓		Ankle jerk p95 (m/s^3) ↓		Obj speed max (m/s) ↓		Foot slip max (m) ↓	
Object	n	Ours	OmniRt	Ours	OmniRt	Ours	OmniRt	Ours	OmniRt	Ours	OmniRt	Ours	OmniRt	Ours	OmniRt	Ours	OmniRt	Ours	OmniRt
box001	6	**0.612**	0.015	**0.821**	0.289	**0.168**	0.607	**0.010**	0.419	**9.1**	17.2	**0.038**	0.507	**540.1**	1315.3	**1.14**	1.95	**0.83**	1.23
box004	3	**0.486**	0.064	0.679	**0.689**	**0.123**	0.571	**0.003**	0.171	**7.2**	13.2	**0.054**	0.541	**805.2**	1555.5	**1.36**	3.51	**1.10**	1.13
box021	11	**0.552**	0.149	0.761	**0.904**	**0.180**	0.579	0.019	**0.001**	14.3	**8.6**	**0.062**	0.225	**707.7**	1638.7	**1.32**	3.54	**1.10**	1.38
box023	7	**0.583**	0.281	0.731	**0.774**	**0.110**	0.337	**0.017**	0.039	20.9	**10.6**	**0.076**	0.272	**739.9**	1743.0	**1.52**	3.05	**0.79**	1.25
box024	4	**0.510**	0.000	**0.860**	0.104	**0.264**	0.751	**0.004**	0.662	**8.3**	16.9	**0.037**	0.526	**613.8**	1611.8	**1.40**	2.28	**1.04**	1.53
bucket003	3	**0.564**	0.009	**0.751**	0.105	**0.250**	0.544	**0.109**	0.462	**15.8**	33.7	**0.038**	0.378	**665.8**	1547.2	**1.36**	2.12	**0.65**	1.52
bucket007	4	**0.492**	0.062	**0.798**	0.240	**0.202**	0.522	**0.013**	0.414	**11.2**	66.9	**0.055**	0.514	**691.5**	1324.3	**1.17**	2.32	**0.60**	1.14
chair006	4	**0.711**	0.038	**0.925**	0.243	**0.173**	0.472	**0.002**	0.344	**7.4**	29.8	**0.045**	0.367	**482.1**	1437.1	**1.08**	1.85	**0.92**	1.16
desk007	3	**0.735**	0.014	**0.862**	0.225	**0.154**	0.842	**0.023**	0.669	**18.2**	45.9	**0.062**	0.814	**527.0**	1181.6	**1.06**	1.56	**0.52**	1.16
desk021	4	**0.619**	0.149	**0.857**	0.627	**0.184**	0.346	**0.002**	0.008	**7.2**	10.7	**0.026**	0.180	**704.7**	1716.8	**1.28**	4.28	**0.89**	1.79
desk023	3	**0.685**	0.101	**0.921**	0.506	**0.170**	0.421	**0.000**	0.144	**6.4**	17.2	**0.044**	0.285	**610.5**	1943.9	**1.31**	3.59	**0.65**	1.36
Overall	52	**0.588**	0.101	**0.804**	0.510	**0.176**	0.536	**0.017**	0.247	**12.3**	21.1	**0.052**	0.381	**653.8**	1560.8	**1.29**	2.83	**0.87**	1.33